IP Address Classes

Class A – 1.0.0.0 to 127 /8

Class B – 128.16.0.0 to 191 /16

Class C – 192.0.0.0 to 223 /24

Class D – 224.0.0.0 to 239 (multicast)

Class E – 240.0.0.0 to 255 (experimental)

Subnetting

Exercises:

IP Address: 198.22.45.173/26

Subnet: 255.255.255.192 (11111111.11111111.11111111.11000000)

IP is class C so 24 bits (2 bits remain)

Network address: 198.22.45.128 (10|101101)

Broadcast address: 198.22.45.191

Valid Hosts range: 198.22.45.129 to 198.22.45.190

IP Address: 172.30.10.130/30

Subnet: 255.255.255.252 (11111111.11111111.11111111.11111100)

IP is class B so 16 bits (14 bits remain)

Network address: 172.30.10.128 (100000|10)

Broadcast address: 172.30.10.131

Valid Hosts range: 172.30.10.129 to 172.30.10.130

IP Address: 192.168.1.0/26

Subnet: 255.255.255.192 (11111111.11111111.11111111.11000000)

IP is class C so 24 bits (2 remain)

Network address: 192.168.1.0 (00|000000)

Broadcast address: 192.168.1.63

Valid Hosts range: 192.168.1.1 to 192.168.1.62

IP Address: 192.168.1.192/26

Subnet: 255.255.255.192 (11111111.11111111.11111111.11000000)

IP is class C so 24 bits (2 remain)

Network address: 192.168.1.192 (11|000000)

Broadcast address: 192.168.1.255

Valid Hosts range: 192.168.1.193 to 192.168.1.254

Network: 192.168.1.0/24 (|00000000)

Broadcast address: 192.168.1.255

Valid Hosts range: 192.168.1.1 to 192.168.1.254

To support 40 devices per network we need 192.168.1.0/26 (00|000000)

4 subnets

64 hosts – 1 network + 1 broadcast = 62 hosts

Network: 176.205.173.188/18 (10|101101.10111100)

Subnet: 255.255.192.0 (11111111.11111111.11000000.00000000)

IP is class B so 16 bits (2 remain)

Network address: 176.205.128.0

Broadcast address: 176.205.191.255

Valid Hosts range: 176.205.128.1 to 176.205.206.254

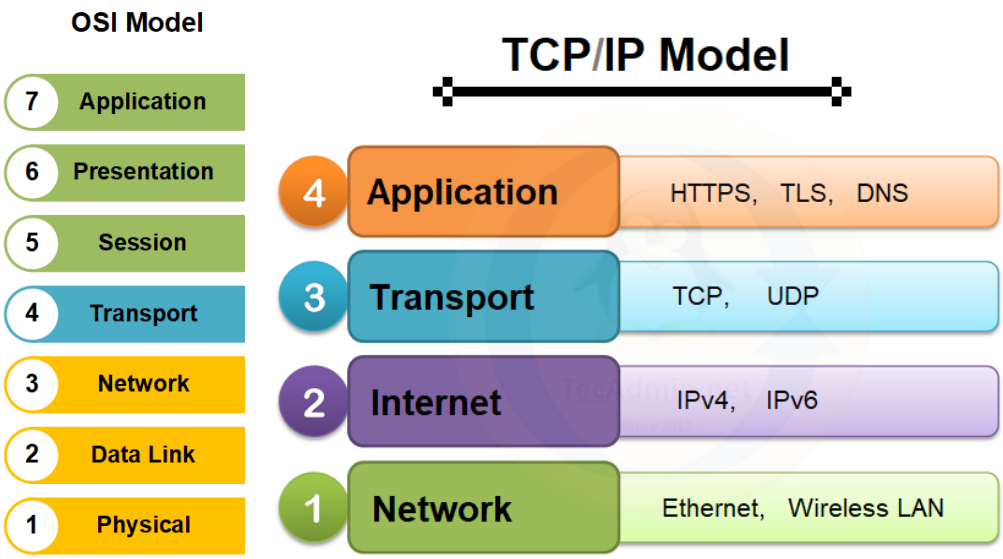
4 subnets

16384 hosts – 1 network + 1 broadcast = 16382 hosts

TPC/IP e OSI

The OSI (Open Systems Interconnection) model is a 7-layer conceptual framework that standardizes how different networking functions interact to enable communication between systems. (Conceptual Model)

The TCP/IP model is a 4-layer networking framework used to describe and standardize internet communication. It maps practical protocols to logical layers that handle addressing, routing, transport, and application services. (Practical and Protocol driven)



OSI:

7. Application: Provides network services directly to end-user applications (e.g., email, web browsers).

6. Presentation: Translates, encrypts/decrypts, and formats data for the application.

5. Session: Manages connections between applications.

4. Transport: End-to-end data delivery, flow control, and error recovery (e.g., TCP, UDP).

3. Network: Logical addressing and routing of packets across multiple networks (e.g., IP, routing).

2. Data Link: Framing, physical addressing (MAC), and error detection/correction on a single link (e.g., Ethernet, PPP).

1. Physical: Transmission and reception of raw bitstreams over physical media (cabling, voltage levels, radios).

TCP/IP:

4. Application: Provides end-user network services and application-level protocols (http, dns, ssh)

3. Transport: End-to-end communication, multiplexing, reliability, and flow control (tcp/udp)

2. Internet: Logical addressing and routing of packets between networks (ipv4/ipv6)

1. Network: Framing, physical addressing (MAC), and sending/receiving frames over a single network link (ethernet, Wi-Fi).

Routers e Switches

A router connects different networks (like your home LAN to the internet WAN) and routes data between them using IP addresses (Layer 3), while a switch connects devices within the same network (LAN) and forwards data using MAC addresses (Layer 2).

Loopback

Testing and Diagnostics: It is used to verify that the computer's internal networking software (the TCP/IP stack) is functioning correctly.

Stability: Unlike physical ports that can "go down" if a cable is unplugged, a virtual loopback interface is always up as long as the device itself is powered on.

Device Identification: Loopback addresses are often used as a stable "Router ID" for routing protocols like OSPF and BGP. Because the address is always active, it ensures the router remains reachable even if one of its physical paths fails.

VLANs

VLANs allow a single physical network switch to be partitioned into multiple separate "virtual switches," each acting as its own distinct broadcast domain.

VLANs primarily operate at **Layer 2** (Data Link Layer) of the OSI model.

Core Purposes and Benefits:

- **Security Isolation:** Sensitive departments (ex: Finance or HR) can be isolated.
- **Performance Optimization:** By breaking a large network into smaller broadcast domains.
- **Cost Efficiency:** Organizations can segment their networks without purchasing additional physical switches or running extra cabling.

How VLANs work:

- **VLAN Tagging (802.1Q):** To distinguish which data belongs to which VLAN, a 4-byte "tag" containing a VLAN ID (a number from 1 to 4094) is added to the Ethernet frame.
- **Access Ports:** Connect to end devices (like PCs or printers). These ports belong to a single VLAN and remove the tag before sending data to the device.
- **Trunk Ports:** Connect switches to other switches or routers. They carry traffic for multiple VLANs simultaneously using tags to identify each frame's destination.
- **Inter-VLAN Routing:** Devices on different VLANs cannot communicate directly. To send data between them, the traffic must be routed through a Layer 3 device, such as a router or a Layer 3 switch.

DNS

The Domain Name System (DNS) is the internet's directory service that translates human-friendly domain names (ex: example.com) into machine-readable IP addresses (ex: 192.0.2.1).

Allows website owners to change hosting providers (changing the IP address) without users needing to learn a new web address.

STP

In networking, STP (Spanning Tree Protocol) is a Layer 2 management protocol designed to prevent loops in Ethernet networks while allowing for physical redundancy.

Without STP, a network with redundant paths would suffer from broadcast storms, where frames circulate endlessly, consuming all available bandwidth and eventually crashing the network.

How it Works (The Election Process):

1. **Elect a Root Bridge:** All switches exchange **BPDUs** (Bridge Protocol Data Units). The switch with the lowest **Bridge ID** (a combination of a priority value and its MAC address) becomes the "Root Bridge"—the central point of the network.

2. **Select Root Ports:** Every non-root switch chooses one "Root Port," which is the port with the lowest-cost path to the Root Bridge.

3. **Designate Ports:** On each network segment, one port is selected as the "Designated Port" to forward traffic.

4. **Block Remaining Ports:** Any ports that are neither Root Ports nor Designated Ports are placed in a **Blocking** (or discarding) state to break potential loops.

DHCP

The Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to automatically assign IP addresses.

The primary goal of DHCP is to eliminate the need for manual network configuration.

NAT

Network Address Translation (NAT) is a networking method used to map an entire private network to one or more public IP addresses by modifying IP header information while transit.

Primary Purposes:

- **IP Address Conservation:** NAT allows thousands of devices on a private network to share a single public IPv4 address, mitigating the global shortage of unique IPv4 addresses.
- **Security and Privacy:** It masks internal IP addresses from the public internet, making it difficult for external attackers to target specific devices directly.
- **Administrative Flexibility:** Organizations can change internal IP schemes or ISPs without needing to readdress every device on the network.

ACLs

An Access Control List (ACL) is a set of rules used to regulate access to digital resources, acting as a "guest list" that specifies which users or systems are permitted or denied entry.

Primary Purposes of ACLs:

- **Security & Traffic Filtering:** ACLs act as a first line of defense on routers and firewalls, inspecting data packets to block malicious or unauthorized traffic.
- **Granular Access Control:** They allow administrators to define precise permissions at the resource level—such as allowing a specific department to access a payroll server while blocking others.

OSPFs

OSPF (Open Shortest Path First) is a standardized link-state routing protocol used to find the most efficient path for data packets within a single "Autonomous System" (a group of networks under one administration).

As of 2026, it remains a dominant protocol in enterprise and data center networks due to its ability to handle large, complex topologies with rapid speed and high reliability.

BGP

BGP (Border Gateway Protocol) is the protocol that manages how the different networks talk to each other. It is effectively the **"Postal Service of the Internet."**

While OSPF (the protocol we discussed earlier) handles the "GPS" work inside a single network, **BGP** (Border Gateway Protocol) is the protocol that manages how those different networks talk to each other. It is effectively the **"Postal Service of the Internet."**

If OSPF is the street map of your city, BGP is the map of the world’s highways that connects one city to another.

What makes BGP different?

1. The Autonomous System (AS)

The Internet isn't one giant network; it is a collection of thousands of smaller networks (ISPs, universities, tech giants like Google). Each of these is called an **Autonomous System (AS)** and is identified by a unique **AS Number (ASN)**. BGP is the protocol used to route traffic between these ASes.

2. Path-Vector vs. Link-State

Unlike OSPF, which uses the Dijkstra algorithm to calculate the best path based on "cost," BGP is a **Path-Vector** protocol. It doesn't look at individual routers; it looks at the sequence of ASes a packet must pass through to reach its destination.

- **Metric:** BGP's primary metric is the **AS_PATH**. It prefers the route that passes through the fewest number of Autonomous Systems.

3. Policy-Based Routing

This is BGP's "superpower." While OSPF always wants the fastest path, BGP allows network admins to choose paths based on **business and political rules**, such as:

- "Don't send traffic through this specific ISP because they are too expensive."
- "Always prefer the link through New York for European traffic, even if the link through Virginia is shorter."

Firewall

Firewall is the **Security Gate** or **Customs Agent** of your network. Its job is to decide who is allowed in and out based on a set of security rules.

How a Firewall Works

At its simplest, a firewall sits between a trusted network (like your office) and an untrusted network (the Internet). It inspects every "packet" of data crossing that boundary and applies a three-step logic:

1. **Inspection:** It looks at the packet's "label" (IP addresses, ports, and protocols).
2. **Comparison:** It checks that information against its **Rule Base** (Access Control List).
3. **Action:** It performs one of three actions:
    - **Allow:** Let the packet through.
    - **Deny:** Block the packet and send a "rejection" message back to the sender.

- **Drop:** Block the packet and say nothing, leaving the sender "hanging."

More subnetting exercises

IP Address: 172.22.148.203 /26

Subnet: 255.255.255.192

Network address: 172.22.148.192 (11|001011)

Broadcast address: 172.22.148.255

Valid Hosts range: 172.22.148.193 - 172.22.148.254

Total usable hosts: 2^6 – 2 = 62